EXP NO-2 :- DETERMINATION OF CURRENT IN CIRCUIT USING MESH ANALYSIS

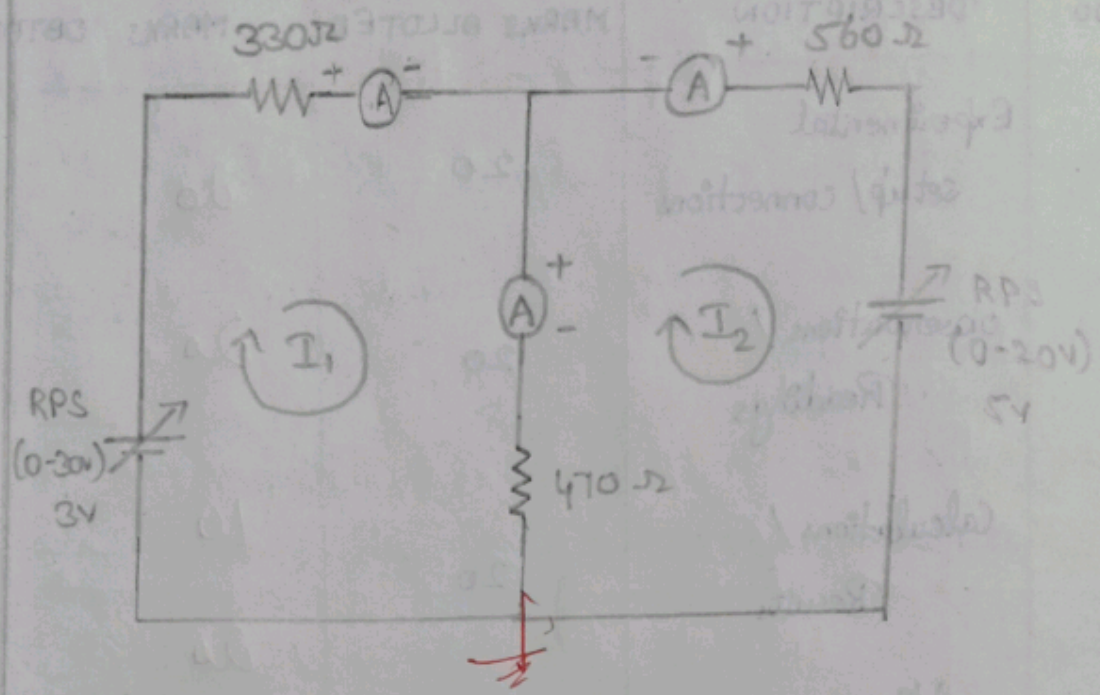
AIM :-

To determine the current using mesh analysis both theoretically and practically for a given DC circuit.

APPARATUS REQUIRED :-

SL.NO	Apparatus	Specifications	Quantity.
1.	Regulated Power supply	(0-30) V	1
2.	Ammeter	(0-10)mA MC	3
3.	Resistor	300Ω, 470Ω, 56Ω	Each,
4.	Bread board	–	1

CIRCUIT DIAGRAM



TABULAR COLUMN:

PARAMETERS	THEORETICAL	PRACTICAL
330Ω	1.2mA	1.3mA
560Ω	4.2mA	4.3mA
470Ω	5.4mA	5.4mA

PROCEDURE :-

1. Give connections as per the circuit diagram

2. Switch ON the supply, vary the RPS and set a particular input voltage.

3. Note down the readings of ammeter and voltmeter and tabulate them.

4. Vary the RPS for different input voltages and note down the readings of all the meters

5. Reduce the RPS to its minimum value and switch OFF the supply.

6. Using the tabulated values, verify Kirchoff's laws practically and verify it theoritically.

CALCULATIONS:

$$-3 + 330I_1 + 470(I_1 - I_2) = 0$$

$$330I_1 + 470I_1 - 470I_2 = 3$$

$$800I_1 - 470I_2 = 3 \rightarrow ①$$

$$560I_2 + 5 + 470(I_2 - I_1) = 0$$

$$560I_2 + 470I_2 - 470I_1 = -5$$

$$-470I_1 + 1030I_2 = 5 \rightarrow ②$$

By Cramer's Rule for ① and ②:

$$\begin{bmatrix} 800 & -470 \\ -470 & 1030 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\Delta = 800(1030) - 470(470)$$

$$\boxed{\Delta = 603100}$$

$$\Delta_1 = \begin{bmatrix} 3 & -470 \\ -5 & 1030 \end{bmatrix}$$

$$= 3090 - 2350$$

$$\boxed{\Delta_1 = 740}$$

$$\Delta_2 = \begin{bmatrix} 800 & 3 \\ -470 & -5 \end{bmatrix}$$

$$= -4000 + 1410$$

$$\boxed{\Delta_2 = -2590}$$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{740}{603100} = 0.0012\ A$$

$$\boxed{I_1 = 1.2\ mA}$$

$$I_2 = \frac{\Delta_2}{\Delta} = \frac{-2590}{603100} = -0.0042\ A$$

$$\boxed{I_2 = -4.2\ mA}$$

$$I_3 = I_1 + I_2$$

$$= 1.4 + 4.2$$

$$\boxed{I_3 = 5.4\ mA}$$

RESULT:-

Thus, the Mesh analysis is verified practically and theoretically. The resultant current for supply are:

a) The current across 330Ω resistor is 1.2MA

b) The current across 560Ω resistor is -4.2MA

c) The current across 470Ω resistor is. 5.4MA

MARKS SPLIT-UP:-

Sl.No	Description.	Marks allocated	Marks obtained
1.	Experimental setup / connection	5	20
2.	Observation / Readings	5	10
3.	Calculation / Result	5	10
4.	Viva.	5	20
5.	Record	5	20
		Total	80